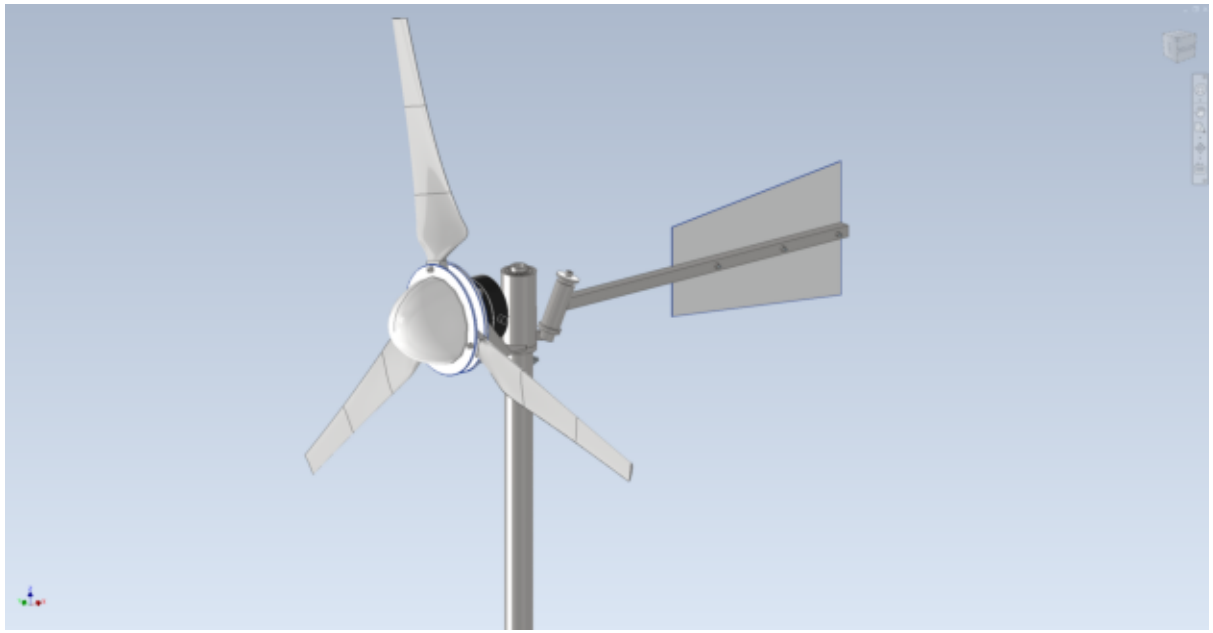
Small Wind Turbine: 1.3 m rotor, 3 printed blades, 400 W PM generator, furling tail

WindTurbine - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / vendor, ASSUMED = n

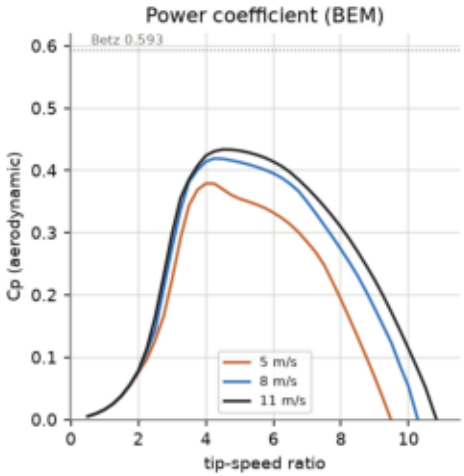


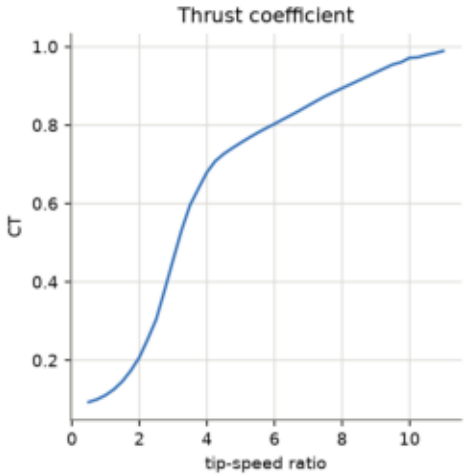
Item	Value	Source
Rotor	D 1300 mm, 3 blades, upwind; SG6040 root -> SG6043 (r 200-400); Schmitz chord 125 -> 47 mm, twist 20.9 -> 4.0 deg for TSR 5	CALC
Aerodynamics	Cp max 0.419 at TSR 4.25 (8 m/s), 0.434 at 11 m/s (BEM + NeuralFoil)	CALC
Generator	ATO-PMG-NE400WM 24 V 3-phase PM, 400 W at 800 rpm, max 445 W, 4.5 kg, $298.97; direct drive	CAT
Output	294 W at 11 m/s (704 rpm, TSR 4.67); 251 W into the battery; max 448 W	CALC
Cut-in / start	charging from 3.6 m/s; standstill start needs 8.5 m/s against the 0.35 Nm catalogue starting torque	CALC
Overspeed	furling tail: starts at 8.5 m/s, 54 deg at 20 m/s; diversion load; shorting brake switch	CALC
Energy	AEP 289 / 398 / 643 kWh at 4.5 / 5 / 6 m/s Rayleigh mean (into the battery)	CALC
Tower	2in Sch 40 tilt-up mast, hub 6.21 m, 4 guys at 4.6 m; worst mast stress 57 MPa (SF 4.2)	CALC
Assembly	67 occurrences, 198 constraints, 0 DOF, 0 clashes at 8 poses; driven by Yaw, Spin, Furl	CAD
Mass	64.1 kg total; head 15.9 kg, rotor 3.25 kg (J 0.098 kg m2), blade 314 g	CAD

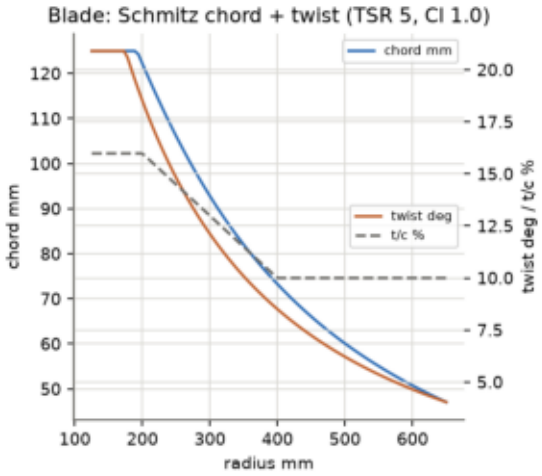
1. Blade design: airfoils, chord and twist, Cp and CT (blade-element momentum)

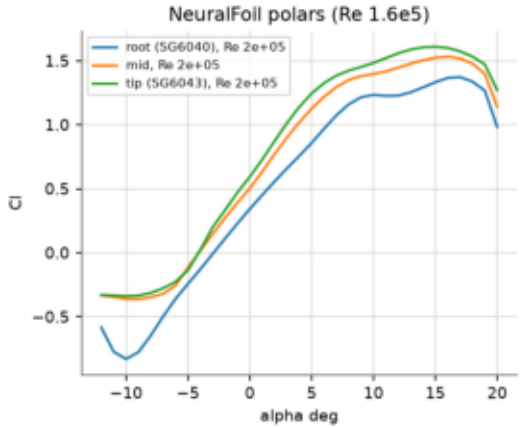
WindTurbine - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / vendor, ASSUMED = n

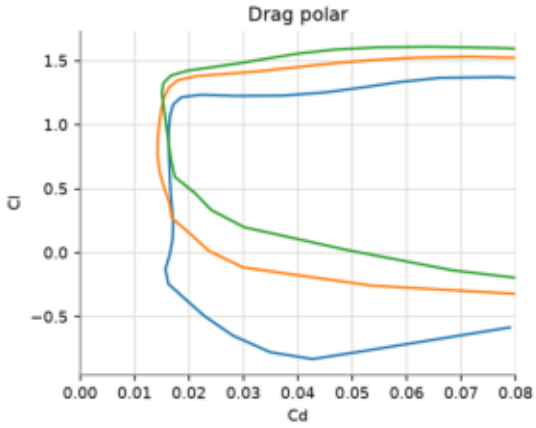
Airfoils: the Selig-Giguere SG604x family was designed for small wind turbines at Re 1e5 - 5e5: SG6040 (16 %) at the root for stiffness and room for the carbon rods, blended linearly to SG6043 (10 %) from r 200 to 400 mm (coordinates from the UIUC database). Polars for every blended section come from NeuralFoil 0.3.3 (a network trained on XFoil) at the local Reynolds number (mean confidence 0.89), extended past 20 deg with Viterna (Cd max 1.23) for start-up and stall. Planform: Schmitz optimum c = 8 pi r (1 - cos phi) / (B Cl), phi = 2/3 atan(1 / lambda_r), with design TSR 5 and Cl 1 (alpha 3.5 deg); chord capped at 125 mm inboard of r 175; twist theta = phi - alpha. The blade is lofted in Inventor through exactly these sections. BEM: Prandtl tip and hub loss, Buhl high-induction correction, pitch 0. Check at the design point (8 m/s, TSR 5): local alpha 3.2 - 5.7 deg and Cl 0.90 - 0.99 along the blade (designed for 3.5 deg / 1.0), axial induction 0.26 - 0.36 (0.33 ideal). Solidity 0.083; Re 113,559 at the root, 130,466 at the tip at 8 m/s. Result: Cp max 0.419 at TSR 4.25 at 8 m/s (0.379 at 5 m/s, Re effect), against 0.593 Betz. The optimum TSR is below the 5.0 design value because the root chord is capped and the tip loss is large on a 3-blade 0.65 m rotor.





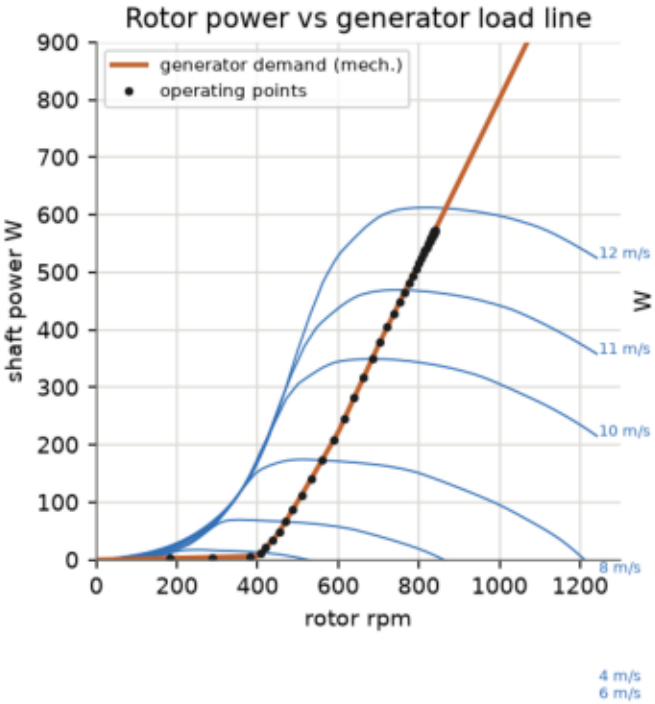


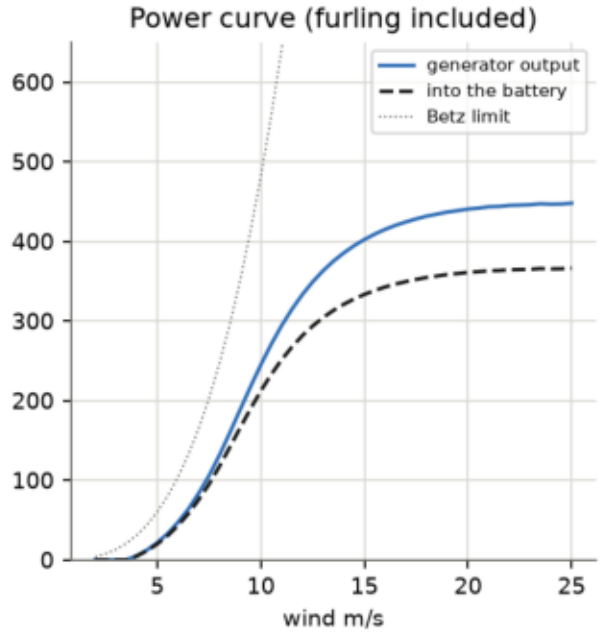


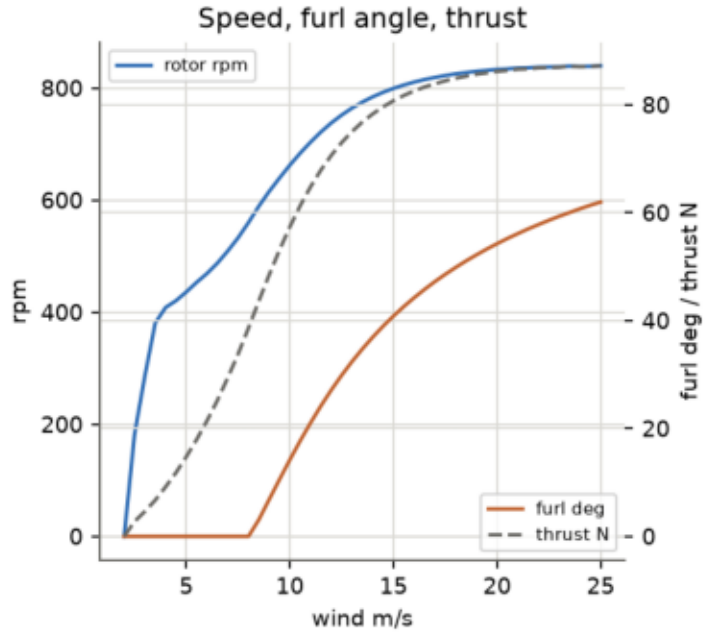


2. Generator matching, power curve and energy

Generator: ATO-PMG-NE400WM, 24 V. The 24 V power curve on the vendor page (an image, digitised by hand) starts charging at 400 rpm and gives 75 / 170 / 290 / 403 W at 500 / 600 / 700 / 800 rpm. I read it as electrical output and assumed 80 % mechanical-to-DC efficiency plus 0.15 Nm running drag (ASSUMED; the catalogue lists only a 0.35 Nm maximur starting torque). Above 800 rpm the curve is extended at its last slope (the furl keeps the rotor below 840 rpm). Operating point at each wind speed: the highest rotor speed where the BEM rotor torque falls through the generator torque (stable intersection). Result: the battery-clamped generator holds the rotor at TSR 4.6 - 5.0 between 6 and 11 m/s, close to the Cp peak, so no MPPT controller is needed. Charging starts at 3.6 m/s (free-running rotor reaches 400 rpm). Output 22 / 132 / 294 / 402 W at 5 / 8 / 11 / 15 m/s; maximum 448 W at 25 m/s, which is at the 445 W catalogue maximum (0.7 % over at 25 m/s). Losses to the battery: 3-phase 12 AWG drop cable 8 m, rectifier 2 x 1.0 V, 2 x 20 m of 8 AWG: 21 W cable loss at 11 m/s, 251 W into the battery. Annual energy into the battery (Rayleigh, hub height, no availability loss): 194 / 289 / 398 / 643 kWh at 4 / 4.5 / 5 / 6 m/s mean. A 6 m tower in a back yard is usually turbulent; treat these as upper bounds.

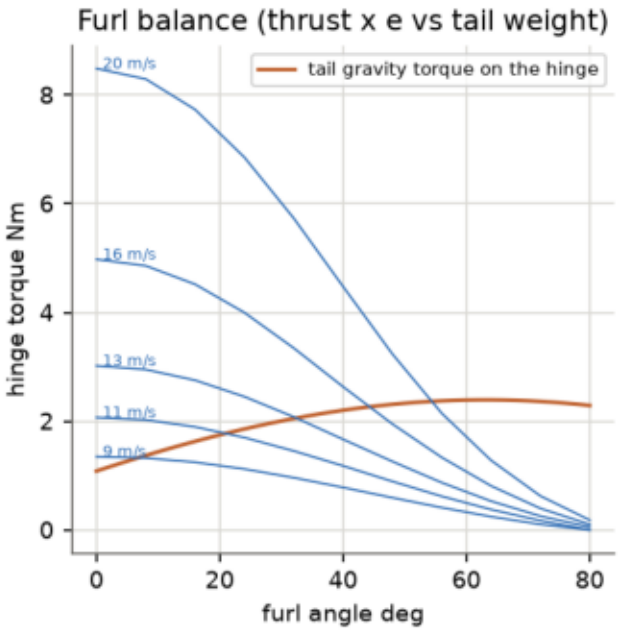


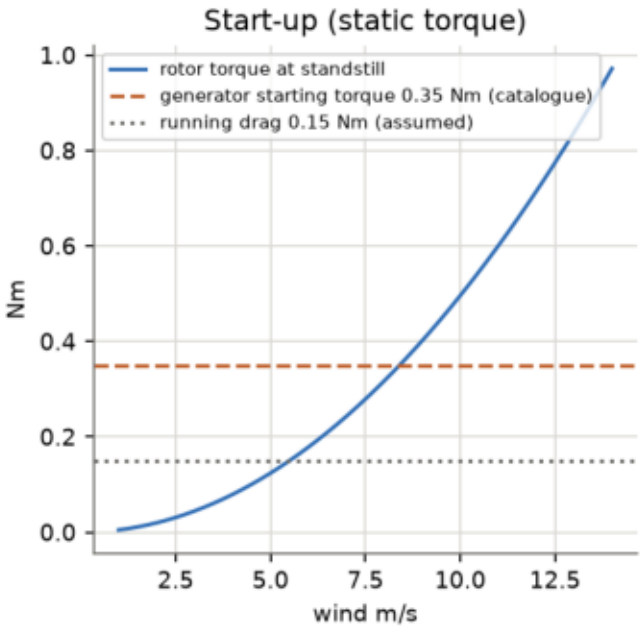


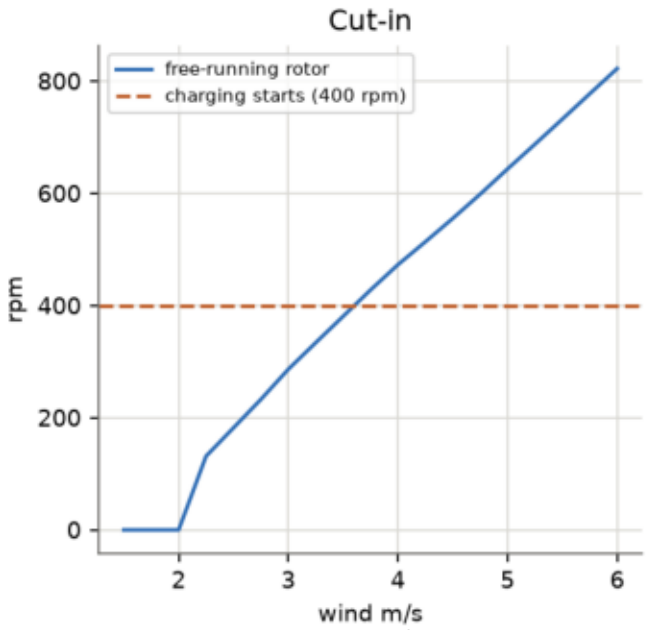


3. Overspeed protection: furling tail, diversion load, brake; start-up and cut-in

Furling: the rotor axis is offset e = 35 mm from the yaw axis, so thrust T makes a yaw moment T e that turns the rotor out of the wind; the tail stays in the wind and swings up its 20 deg inclined hinge. Its weight resists: tail mass 2.02 kg with CG [352.26, 3.18, 114.29] mm in the hinge frame (CAD). The gravity torque on the hinge, dU/dphi from the CAD CG rotated about the hinge axis, is 1.08 Nm on the rest stop and 2.29 Nm at 80 deg. The yaw moment reaches the hinge as T e x L_hinge cos(delta) / L_yaw = 0.784 T e (vane centre of pressure 0.610 m from the yaw axis). Quasi-static balance with the rotor seeing V cos(phi): furling starts at 40 N thrust = 8.5 m/s, 41 / 54 / 62 deg at 15 / 20 / 25 m/s, and the output levels off at about 440 W. A first design (hinge 30 deg, rest 40 deg) let the output reach 810 W because its tail torque rose too steeply; a sweep of 20 tilt / rest pairs picked 20 / 25 deg. Tail aerodynamics, gyroscopic moments and yaw dynamics are not modelled (quasi-static), so the furl speed needs a field check.
Runaway (controller or battery disconnected, generator unloaded, furl still active): 1573 rpm maximum, used as the design overspeed for the blade root. Diversion load: TriStar TS-45 in diversion mode holds the bank at 28.8 V and switches the surplus into 2 x 0.6 ohm 500 W resistors: 24 A / 691 W capacity vs 15.6 A maximum turbine current (x1.54).
Brake: shorting the three phases. With a DC-equivalent model fitted to the curve (E = k n (k 0.0675 V/rpm DC equivalent from the 400 rpm charge start), R_dc = 1.81 ohm fitted at 800 rpm) the shorted generator holds the rotor at 10.0 rpm at 10.0 m/s, 19.0 rpm at 12.0 m/s, 38.0 rpm at 15.0 m/s: a service stop, not a storm brake (inductance is neglected, so the real torque at speed is lower).
Start-up: the rotor torque at standstill (every section near 70-85 deg angle of attack) reaches the 0.35 Nm catalogue starting torque only at 8.5 m/s, and the 0.15 Nm running drag at 5.6 m/s. Cogging is periodic, so gusts usually start a rotor earlier than this static number, but it is the main open item: measure the breakaway torque of the delivered generator.

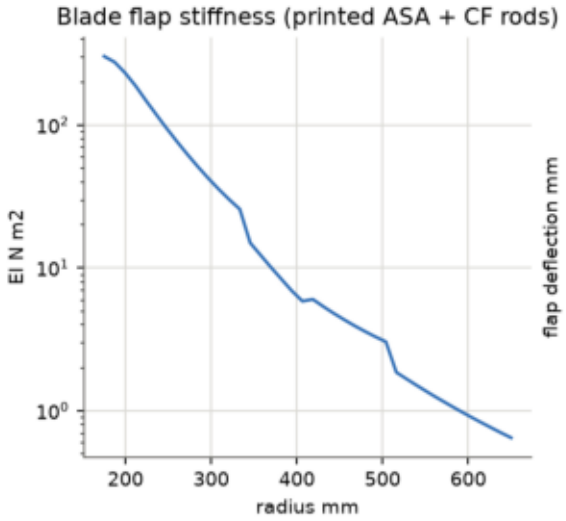


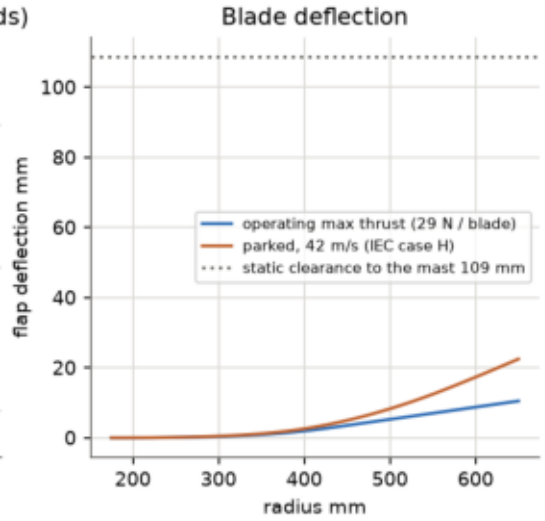


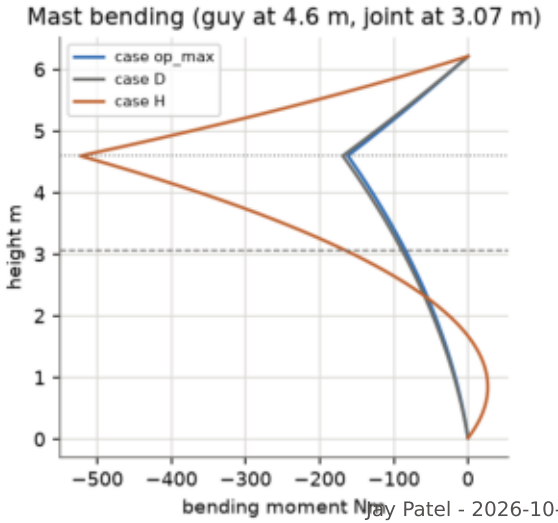


4. Loads (IEC 61400-2 simplified load model), blade structure, tower

Check	Result	Source
Load cases	IEC 61400-2 class IV: V_ref 30, V_ave 6, V_e50 42 m/s. Case D max thrust (C_T 0.5, 2.5 V_ave): 92 N. Case H parked at V_e50 (Cd 1.5 on the blade planform): 69 N per blade, 208 N rotor + 60 N nacelle. Max operating thrust 87 N (furl-limited).	CALC
Blade root (printed ASA block 44 x 20 at the clamp edge r 120)	M 15.2 Nm (case H) + centrifugal 2232 N at 1573 rpm runaway, combined: 7.7 MPa vs 15 MPa across printed layers: SF 1.94	CALC
Blade joints (carbon rods carry all bending)	r 270: 2 x 5 mm rods, 6.9 Nm, 282 MPa vs 1400: SF 5.0; r 460: 2 x 3 mm, 1.5 Nm, 287 MPa: SF 4.9; epoxy bond shear 0.75 / 0.84 MPa at runaway. Rod walls >= 1.1 mm (design.py check)	CALC
Root bolts and clamp	2 x M8 per blade: 7.0 MPa bearing on the printed block at runaway; 6 x M8 A2 at 12 Nm clamp friction 9000 N per blade pair of faces	CALC
Deflection and frequency	tip 10 mm at max operating thrust, 22 mm parked case H (E 1.36 GPa printed ASA + rods) vs 109 mm blade-to-mast clearance (CAD). Flap mode about 23 Hz (estimate) - crossed by 3P at about 450 rpm	CALC/CAD
Mast (2in Sch 40 A53 B, guyed at 4.6 m)	case H: 268 N at the hub + mast drag; 523 Nm at the guy point, 57 MPa: SF 4.2 on 241 MPa; thread root at the coupling SF 7.1	CALC
Guys, anchors, buckling	guy angle 49 deg, length 6.1 m, pretension 400 N, worst 1436 N: SF 13 on 3/16in cable, 7.7 on a 2500 lb screw anchor (ASSUMED rating); mast axial 2524 N vs Euler 26090 N: SF 10	CALC
Tower-top frequency	about 7 Hz (head on the 1.6 m overhang above the guys, rough) - inside the 1P band 6.7-14.0 Hz: balance the rotor, check by a tap test; guying at 5.2 m moves it to about 14 Hz	CALC







5. Bearings, yaw spindle, hub, electrical system

Item	Result	Source
Yaw bearings 2 x 6007-2RS (148 mm apart)	case H moment couple: 423 N lower, 155 N upper radial + 156 N weight; static s0 20 (C0 10.2 kN). Yaw motion is slow oscillation: static check governs	CALC/CAT
Yaw spindle 35 / 22 (1045) at the collar	25 Nm (head force + weight offset -18 mm): 7 MPa, SF 44	CALC
Generator bearings (inside the ATO unit)	6203-2RS front / rear (17 mm shaft), C 9.56 kN, C0 4.75 kN; front overhang 55 mm, span 70 mm (assumed): front Fr 57 N, Fa 65 N at 11 m/s, P 149 N; Rayleigh-5 m/s weighted L10 about 5519 years; parked case H s0 34. Bearing type not published	CALC/ASSUMED
Hub on the 17 mm shaft	M16x1.5 nyloc at 60 Nm: 18750 N preload, friction torque 59 Nm vs 6.5 Nm peak rotor torque: SF 9; M8 x 10 engaged in 6061-T6: ~11 kN shear-out vs 1116 N per bolt (centrifugal / 2)	CALC
Electrical	3-phase AC down the mast (3 x 12 AWG, 8 m) to a stop switch and MDS100A rectifier at the base, 2 x 8 AWG 20 m to the 24 V AGM bank (2 x Renogy 100 Ah); TriStar TS-45 diversion control with a 1.2 ohm dump load; 40 A breaker + 60 A fuse; ground rod + DC surge arrester. AGM rather than lithium: a BMS that opens would unload the turbine into runaway	design
Wiring loss	21 W at 11 m/s (7 %) - acceptable; move the bank closer or use 6 AWG to halve it	CALC

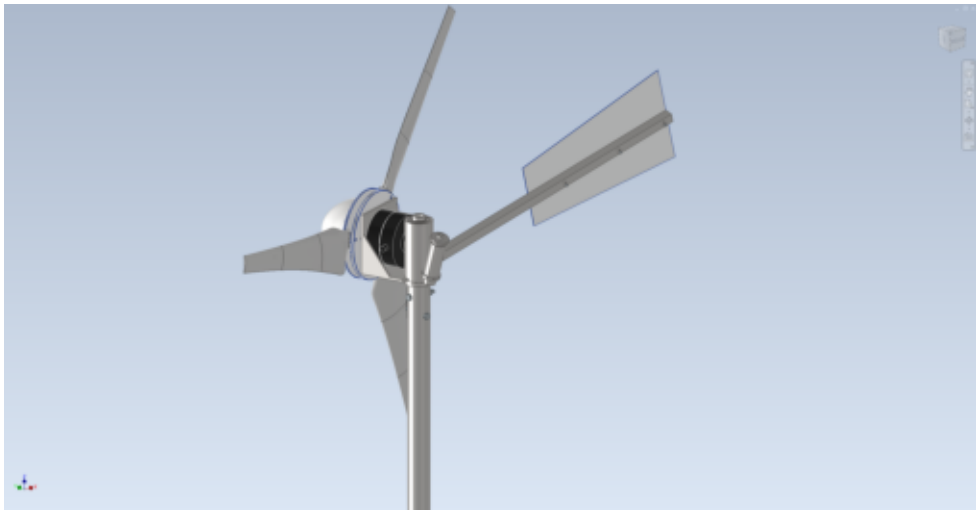
Assumptions: air density 1.225 kg/m3; generator efficiency 80 % and drag 0.15 Nm; printed ASA 1.36 GPa effective modulus, 15 MPa across layers / 30 MPa along; pultruded carbon 1400 MPa flexural; guy anchor 2500 lb; generator internal bearings 6203; IEC class IV site. Not modelled: yaw dynamics and gyroscopic loads (IEC case B), fatigue (IEC case A), tower-shadow and turbulence, icing, blade erosion. Prices: generator, TS-45 controller, ASA filament and 3 mm carbon rod confirmed on vendor pages (2026-10-29); everything else estimated.

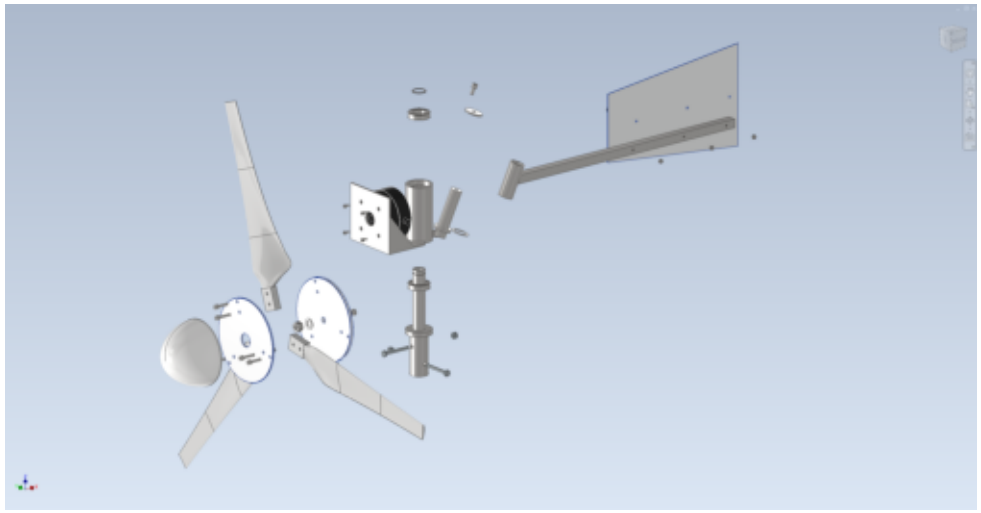
6. Inventor model: working-model check, clearances, pictures

WindTurbine - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / vendor, ASSUMED = n

Pose	Occ.	Clashes	mm3	Probe errors (deg)
design: Yaw 0 / Spin 0 / Furl 0	67	0	0	NacelleFrame 0.00000, Hub 0.00000, BladeTip 0.00000, TailVane 0.00000
Spin 60 (blade down, at the tower)	67	0	0	NacelleFrame 0.00000, Hub 0.00000, BladeTip 0.00000, TailVane 0.00000
Spin 100	67	0	0	NacelleFrame 0.00000, Hub 0.00000, BladeTip 0.00000, TailVane 0.00000
Furl 40 / Spin 25	67	0	0	NacelleFrame 0.00000, Hub 0.00000, BladeTip 0.00000, TailVane 0.00013
Furl 80 (full) / Spin 60	67	0	0	NacelleFrame 0.00000, Hub 0.00000, BladeTip 0.00000, TailVane 0.00005
Furl 80 / Spin 0	67	0	0	NacelleFrame 0.00000, Hub 0.00000, BladeTip 0.00000, TailVane 0.00005
Yaw 90 / Spin 20 / Furl 80	67	0	0	NacelleFrame 0.00000
Yaw -150 / Spin 100 / Furl 25	67	0	0	NacelleFrame 0.00000

CAD check	Value
DOF / unhealthy	0 / 0
Blade - mast (over a turn)	108.8 mm
Blade - generator plate	14.0 mm (root, stiff)
Tail vane - blades, full furl	280 mm
Tail - mast, full furl	57 mm
Hub - generator plate	2.0 mm
Envelope	1252 x 1158 x 6865 mm





Left: furled at Yaw -70 / Furl 70 (tail stays downwind). Right: exploded head. Pictures captured in Inventor.